Name: ______________________________ Date: ____________________

Physics 438A Group Activity #6

Consider the dynamical gauge transformation discussed in Lecture #3, Section 3.1. Given a time-dependent unitary operator $V(t)$, how must the time evolution operator $U(t, t_0)$ transform to maintain the unitary time evolution rule?

$$|\psi(t)\rangle = U(t, t_0)|\psi(t_0)\rangle \quad \mapsto \quad |\tilde{\psi}(t)\rangle = \tilde{U}(t, t_0)|\tilde{\psi}(t_0)\rangle$$